Example 6.35 Find the equation of the pair of lines through the origin and perpendicular to the pair of lines $ax^2 + 2hxy + by^2 = 0$

Solution:

Let m_1 and m_2 be the slopes of these two lines.

$$y - m_1x = 0 \text{ and } y - m_2x = 0 \tag{6.31}$$

Combined equation of these two lines is

$$\begin{aligned} (y - m_1x)(y - m_2x) &= 0 \\ y^2 - (m_1 + m_2)xy + m_1m_2x^2 &= 0 \end{aligned} \tag{6.32}$$

Given that

$$ax^2 + 2hxy + by^2 = 0 \tag{6.33}$$

$$\text{Thus, } m_1 + m_2 = \frac{-2h}{b} \quad \text{and} \quad m_1m_2 = \frac{a}{b} \tag{6.34}$$

The lines perpendicular to (6.31) are

$$y + \frac{1}{m_1}x = 0 \quad \text{and} \quad y + \frac{1}{m_2}x = 0$$

The combined equation is

$$\begin{aligned} (m_1y + x)(m_2y + x) &= 0 \\ m_1m_2y^2 + (m_1 + m_2)xy + x^2 &= 0 \end{aligned}$$

By using (6.34) $\quad \frac{a}{b}y^2 - \frac{2h}{b}xy + x^2 = 0$

$$\text{The required equation is } ay^2 - 2hxy + bx^2 = 0 \tag{6.35}$$

6.5.3 Equation of the bisectors of the angle between the lines $ax^2 + 2hxy + by^2 = 0$

Let the equations of the two straight lines be $y - m_1x = 0$ and $y - m_2x = 0$

$$\therefore \ m_1 + m_2 = -\frac{2h}{b} \quad \text{and} \quad m_1m_2 = \frac{a}{b}$$

We know that the equation of bisectors is the locus of points from which the perpendicular drawn to the two straight lines are equal.

Let $P(p, q)$ be any point on the locus of bisectors.

The perpendiculars from $P(p, q)$ to the line $y - m_1x = 0$ is equal to the perpendicular from $P(p, q)$ to $y - m_2x = 0$

$$\pm\frac{q - m_1p}{\sqrt{1 + m_1^2}} = \pm\frac{q - m_2p}{\sqrt{1 + m_2^2}}$$

$$\text{That is,} \qquad (q - m_1p)^2\left(1 + m_2^2\right) = (q - m_2p)^2\left(1 + m_1^2\right)$$